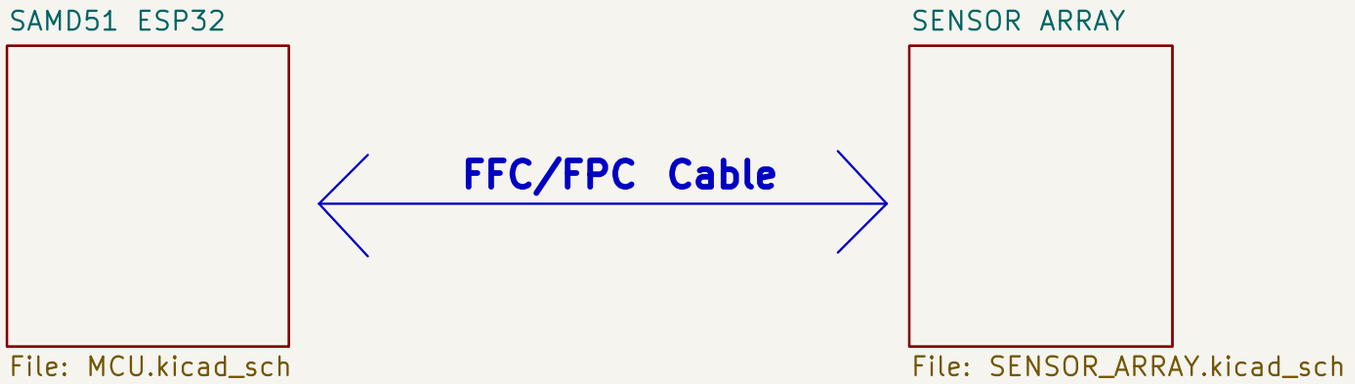


Sheet: /
File: SAVITAR_V2.kicad_sch

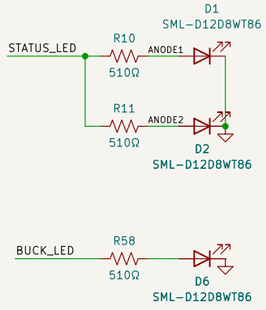
Title:

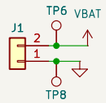
Size: User	Date:	**Rev:**
KiCad E.D.A. 10.0.0		Id: 1/3

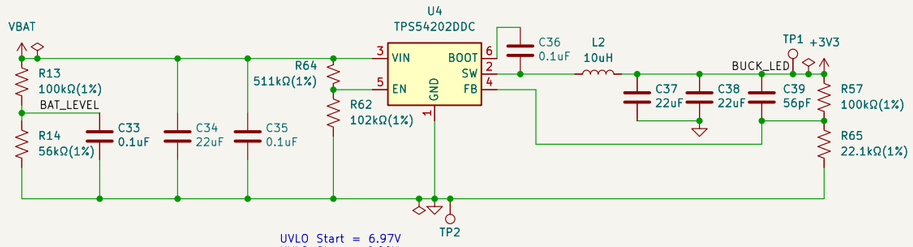
CHECKLIST
- What footprint for SWD connection circuit? OK
- Problem with duplicate footprint for same designator

PUT THE DAMN ORDER ALREADY!?!?

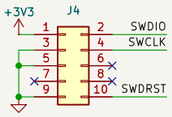
STATUS LEDs



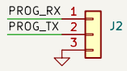




SAMD51 SWD PROGRAMMING

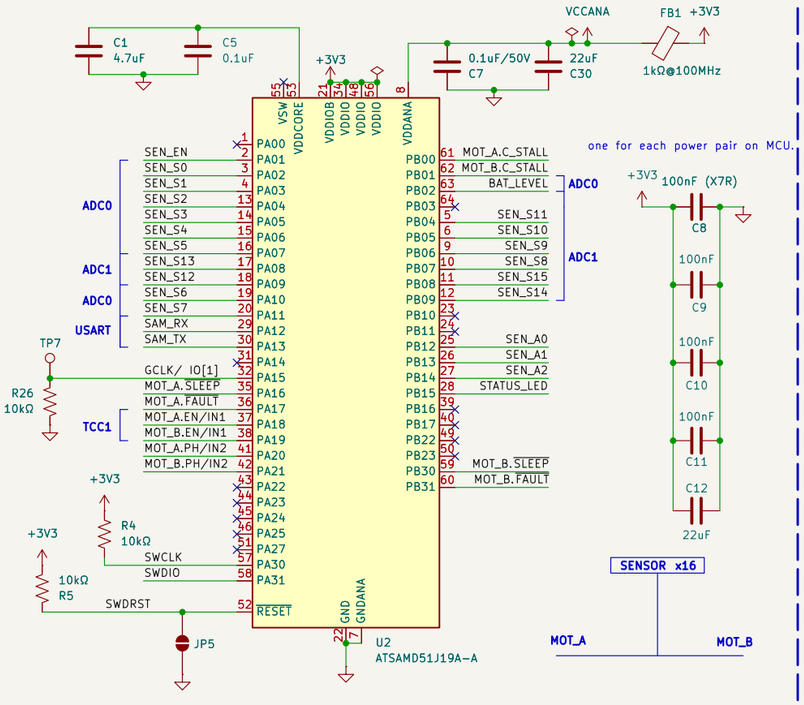


ESP32 UART PROGRAMMING



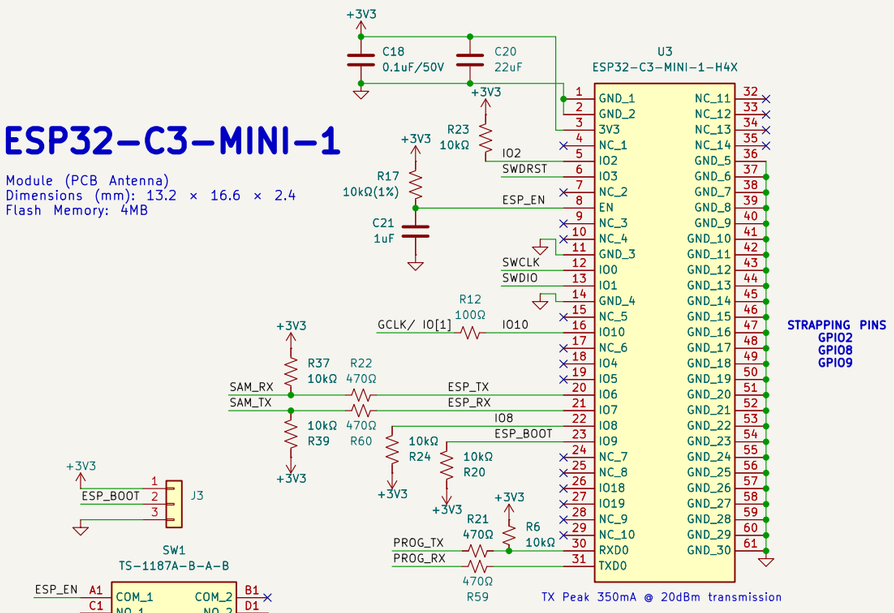
ATSAMD51J19A

TQFP-64
Dimensions (mm): 12 × 12 × 1.2
Program Memory: 512KB
Data Memory: 192KB



ESP32-C3-MINI-1

Module (PCB Antenna)
Dimensions (mm): 13.2 × 16.6 × 2.4
Flash Memory: 4MB

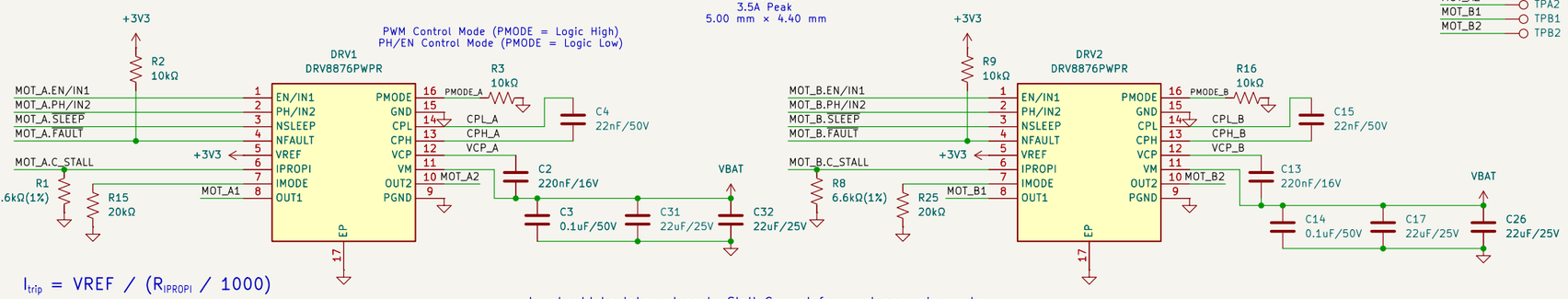


H7 MountingHole
H8 MountingHole

DRV8876 DRIVERS

HTSSOP-16
3.5A Peak
5.00 mm × 4.40 mm

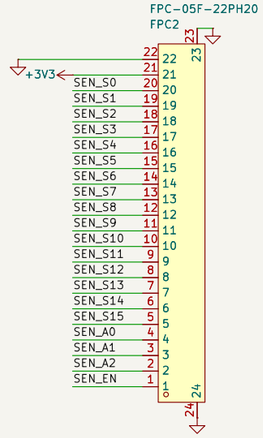
PWM Control Mode (PMODE = Logic High)
PH/EN Control Mode (PMODE = Logic Low)



I_{trip} = VREF / (R_{IPROPI} / 1000)

I_{trip} should be taken close to Stall Current for most accurate reads.

SENSOR ARRAY CONNECTOR

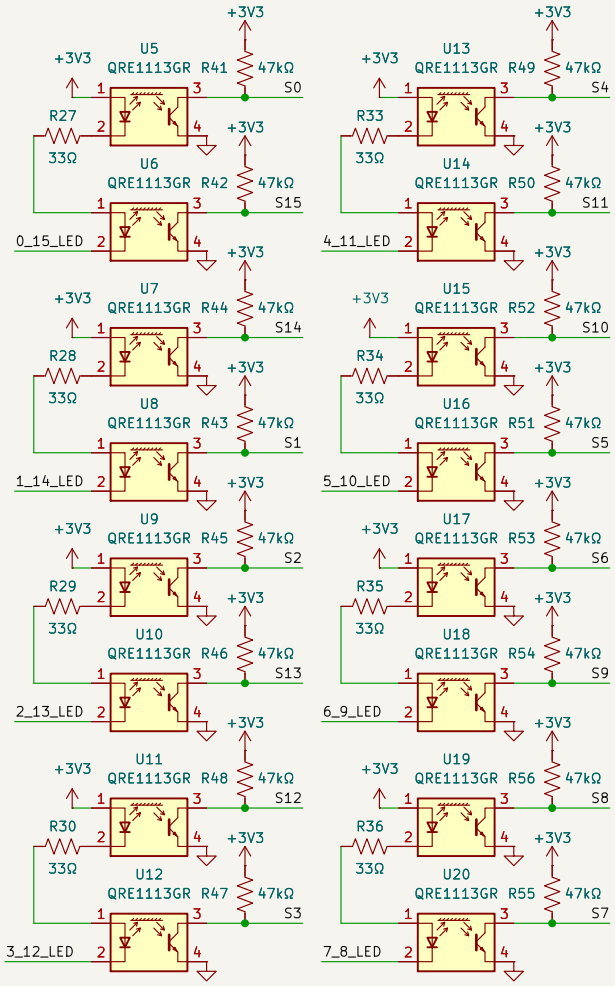


Sheet: /SAMD51 ESP32/
File: MCU.kicad_sch

Title: SAMD51 ESP32 MCU

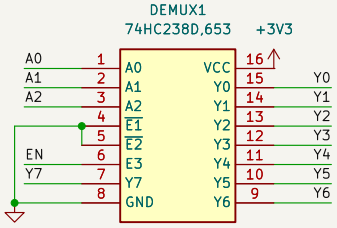
Size: A3	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 2/3

SENSOR ARRAY

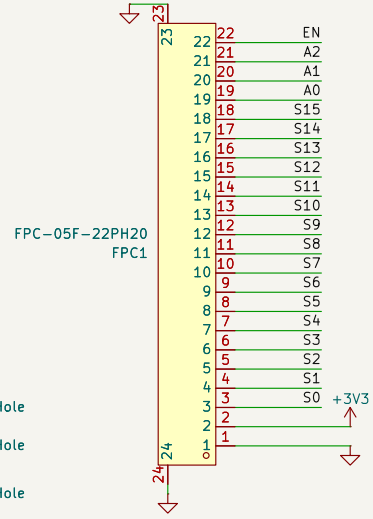


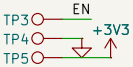
33Ω will give 24mA vs 47Ω that will give 17mA to the emitter.

POWER PAIR

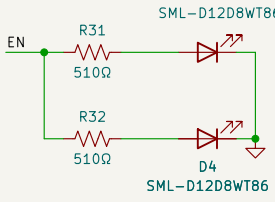




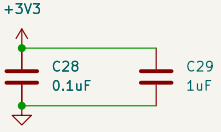








STATUS LED



Each sensor of MUX1 and MUX2 is powered and read in parallel.

READING SEQUENCE

SELECT ADDR 0 |-> NMOS 0 Triggered -> LED 0 & LED 15 ON -> OFF
|-> WAIT TILL STABLE -> READ ADC0 & ADC1

Alexandru Moț

Sheet: /SENSOR ARRAY/
File: SENSOR_ARRAY.kicad_sch

Title: SAVITAR Version 2

Size: User	Date:	Rev:
KiCad E.D.A. 10.0.0		Id: 3/3